LAB 4: Subnetting and Supernetting Using Cisco Packet Tracer

OBJECTIVE:

1. To understand logical division of IP networks through **subnetting**.
2. To understand aggregation of networks through **supernetting (CIDR)**.
3. To design and implement subnetting and supernetting topologies in **Cisco Packet Tracer**.
4. To verify communication between different subnets using **ping**.

THEORY:

1. Subnetting

Subnetting is the process of dividing a single large network into multiple smaller logical networks called **subnets**. It is achieved by borrowing bits from the host portion of an IP address to create additional network identifiers.

Operation:
A subnet mask is used to extend the network portion of the address. For example, borrowing 2 bits from a Class C /24 network results in 4 subnets. Routers use the subnet mask to determine which subnet an IP address belongs to.

Application:
Subnetting improves network performance and security by reducing broadcast traffic and allowing efficient IP address utilization.

2. Supernetting

Supernetting, also known as **Classless Inter-Domain Routing (CIDR)**, is the process of combining multiple smaller networks into a single larger network.

Operation:
The subnet mask is modified by moving the network boundary to the left, allowing multiple contiguous networks to be summarized into one routing entry.

Application:
Supernetting reduces routing table size and improves routing efficiency in large-scale networks.

NETWORK DESIGN:

1. Subnetting Design

Base Network: 192.168.1.0 /24
Required Subnets: 4
Borrowed Bits: 2
New Subnet Mask: /26 → 255.255.255.192
Block Size: 64 IP addresses per subnet

Subnet Calculation Table

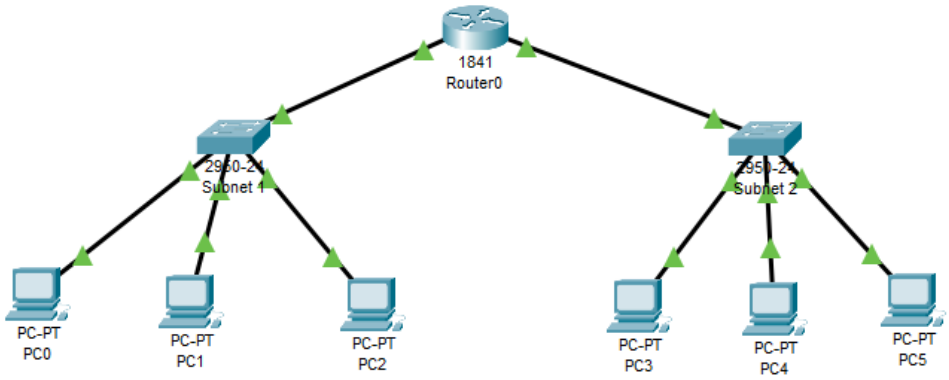
Subnet	Network ID	1st Host	Last Host	Broadcast ID
1	192.168.1.0	192.168.1.1	192.168.1.62	192.168.1.63
2	192.168.1.64	192.168.1.65	192.168.1.126	192.168.1.127
3	192.168.1.128	192.168.1.129	192.168.1.190	192.168.1.191
4	192.168.1.192	192.168.1.193	192.168.1.254	192.168.1.255

Subnetting Topology

A router is used to connect two subnets:

- Subnet 1: 192.168.1.0 /26
- Subnet 2: 192.168.1.64 /26

Switch0 connects PCs (PC0, PC1, PC2) in Subnet 1.
Switch1 connects PCs (PC3, PC4, PC5) in Subnet 2.
Both switches are connected to Router0.



Subnetting IP Configuration Table

Device	IPv4 Address	Subnet Mask	Default Gateway
Router0 (Fa0/0)	192.168.1.1	255.255.255.192	N/A
Router0 (Fa0/1)	192.168.1.65	255.255.255.192	N/A
PC0	192.168.1.2	255.255.255.192	192.168.1.1
PC1	192.168.1.3	255.255.255.192	192.168.1.1
PC2	192.168.1.4	255.255.255.192	192.168.1.1
PC3	192.168.1.66	255.255.255.192	192.168.1.65
PC4	192.168.1.67	255.255.255.192	192.168.1.65
PC5	192.168.1.68	255.255.255.192	192.168.1.65

Ping Verification – Subnetting

- Ping from **PC0** → **PC2** (Same subnet) → Successful

```
C:\>ping 192.168.1.4

Pinging 192.168.1.4 with 32 bytes of data:

Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

- Ping from **PC0** → **PC5** (Different subnet via router) → Successful

```
C:\>ping 192.168.1.68

Pinging 192.168.1.68 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.68: bytes=32 time<1ms TTL=127
Reply from 192.168.1.68: bytes=32 time=1ms TTL=127
Reply from 192.168.1.68: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.1.68:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

2. Supernetting Design

Supernet Network

To combine four Class C networks:

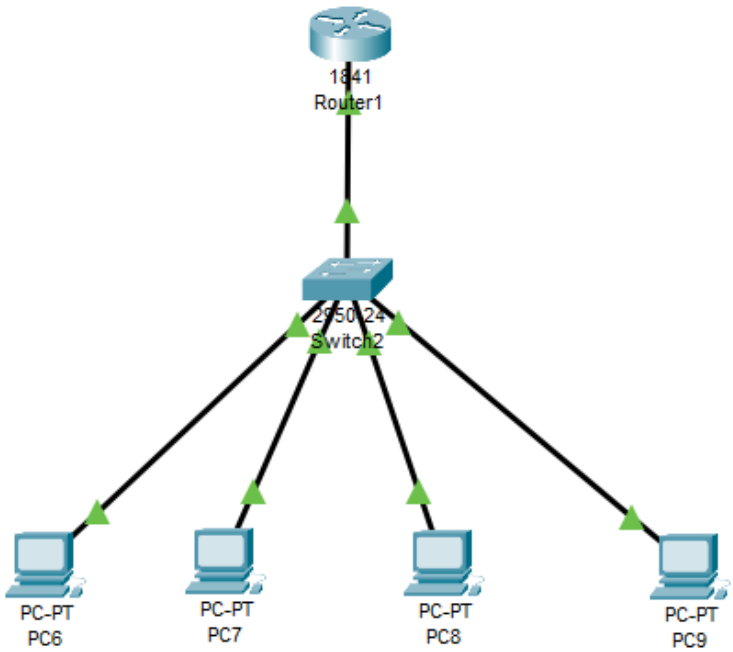
- Supernet Address: **192.168.0.0 /22**
- Subnet Mask: **255.255.252.0**

This single supernet includes:

- 192.168.0.0
- 192.168.1.0
- 192.168.2.0
- 192.168.3.0

Supernetting Topology

One router connects a switch, which connects four PCs (PC6–PC9).
All PCs belong to different networks but share the same supernet mask.



Supernetting IP Configuration Table

Device	IPv4 Address	Subnet Mask	Default Gateway
Router0 (Fa0/0)	192.168.0.1	255.255.252.0	N/A
PC6	192.168.0.10	255.255.252.0	192.168.0.1
PC7	192.168.1.10	255.255.252.0	192.168.0.1
PC8	192.168.2.10	255.255.252.0	192.168.0.1
PC9	192.168.3.10	255.255.252.0	192.168.0.1

Ping Verification – Supernetting

- Ping from **PC6 → PC9** → Successful

```
C:\>ping 192.168.3.10

Pinging 192.168.3.10 with 32 bytes of data:

Reply from 192.168.3.10: bytes=32 time=1ms TTL=128
Reply from 192.168.3.10: bytes=32 time<1ms TTL=128
Reply from 192.168.3.10: bytes=32 time<1ms TTL=128
Reply from 192.168.3.10: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.3.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

RESULT:

The networks were successfully configured.
Ping tests confirmed:

- Communication within the same subnet
- Communication across different subnets via router
- Successful connectivity in the supernetted network

This verifies correct implementation of both subnetting and supernetting.

DISCUSSION & CONCLUSION:

In this lab, subnetting and supernetting were implemented to understand efficient IP address management and routing design. Subnetting allowed division of a larger network into smaller logical networks, improving address utilization and reducing broadcast traffic. Supernetting enabled multiple networks to be summarized into a single routing entry, simplifying routing tables.

Overall, the lab provided practical understanding of IP addressing, subnet mask calculation, router configuration, and packet flow verification using Cisco Packet Tracer.